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TADST: reconstruction with spatio-temporal feature fusion for deviation-based time series anomaly detection

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Abstract

Anomaly detection is crucial in time series analysis for identifying abnormal events. To address the limitations of traditional methods in integrating spatiotemporal correlations and modeling normal patterns, we propose a Time Series Anomaly Detection Model Based on Spatio-Temporal Feature Fusion (TADST). First, the Spatio-Temporal Feature Fusion Network (STF) combines temporal convolutional networks and graph attention influence networks to capture temporal dynamic dependencies and attribute correlations respectively, facilitating joint spatiotemporal feature modeling. Then, the Time Series Reconstruction Network (TSR) employs a multi-layer encoder-decoder architecture to learn the normal sample distribution and amplify discrepancies between reconstructed and anomalous data. Finally, the Anomaly Detection Mechanism (ADM) identifies anomalies by fitting the tail distribution of reconstruction deviations. When the anomaly score exceeds a predefined threshold, the mechanism updates the parameters of the Generalized Pareto Distribution, keeping the detection criteria adaptive. Experiments demonstrate that the proposed TADST achieves state-of-the-art results on five publicly available datasets.

Keywords Artificial intelligence · Deep learning · Anomaly detection · Time series · Representation learning

1 Introduction

Time series are essential in fields such as economics and meteorology, where analyzing these data can reveal key trends and cyclical patterns, supporting effective decision-making [1]. However, sensor failures, data entry errors, and other factors may lead to point anomalies, contextual anomalies, and collective anomalies. Moreover, the unpredictability and high variability of anomalies make it challenging for anomaly detection models to generalize well, often resulting in poor accuracy [2]. Traditional methods, such as those based on linear models, distance metrics, and support vector machines, are widely used but often struggle to handle the complexities and nonlinearities inherent in real-world scenarios.

With the rapid development of artificial intelligence technologies, time series anomaly detection methods are broadly classified into two categories: reconstruction-based and prediction-based methods [3]. Reconstruction-based methods typically adopt an encoder-decoder architecture, where the encoder maps the input sequence into feature space to obtain hidden representation, and the decoder reconstructs the sequence based on the encoded feature. As shown in Fig. 1, if the deviations between the reconstructed and the original data points are significant, they are identified as anomalies. For example, Sakurada et al. [4] pointed out that autoencoders can detect subtle anomalies that linear PCA fails to identify. Moreover, by extending to denoising autoencoders, accuracy can be improved without the need for the complex computations required by kernel PCA.

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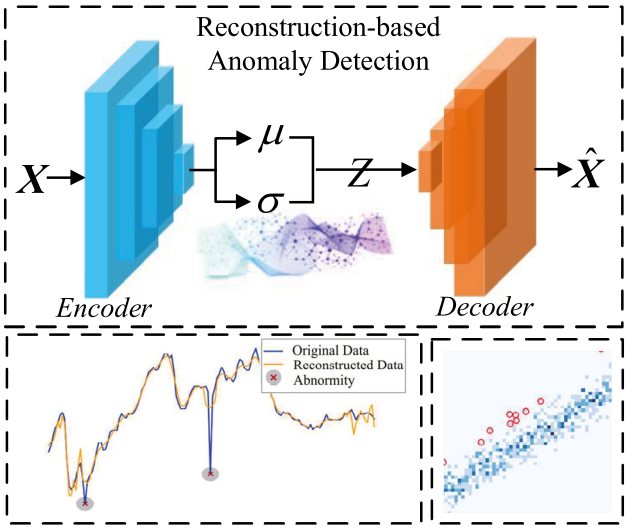


Fig. 1 Time Series Anomaly Detection Based on Reconstruction

Zhang et al. [5] proposed AMSL based on convolutional autoencoders to learn both generic and specific features. This approach leverages an adaptive memory fusion network to integrate global and local memory, and incorporates a self-supervised learning framework to extract general patterns from normal data. On the other hand, prediction-based methods utilize historical data to forecast future time series and compare these predictions with actual observations. When there is a significant deviation between the two, the sample is flagged as an anomaly.

Despite the effectiveness of these methods, they still face three major challenges: (***i***) When modeling the normal sample distribution, the traditional reconstruction-based methods often capture non-representative features, making them ineffective in detecting genuine anomalous patterns. (***ii***) Deep encoding layers may generate redundancies and noise, which interfere with reconstruction during decoding, increasing the risk of overfitting and weakening generalization capability. (***iii***) Due to the dynamic nature of time series, relying solely on the fixed threshold for anomaly detection is insufficient to address the complexity and variability inherent in real-world environments. Additionally, common challenges such as class imbalance, missing values, irregular sampling, and model interpretability also remain.

To address the above issues, we introduce a multivariate time series anomaly detection model, TADST. As shown in Fig. 4, the Spatio-Temporal Feature Fusion Network analyzes the spatiotemporal interactions between temporal and attribute dimensions, providing rich feature representation for subsequent reconstruction. Then, the Time Series Reconstruction Network learns the normal sample distribution of sequences, amplifying the discrepancies between potential anomalies and normal data while fitting non-anomalous data segments. Finally, the Anomaly Detection Mechanism determines whether a data point is anomalous based on its reconstruction deviation, labeling any point whose anomaly score exceeds the corresponding threshold as an anomaly, which also triggers an update to the threshold.

The main contributions of this paper are as follows:

- We propose a Time Series Anomaly Detection Model Based on Spatio-Temporal Feature Fusion (TADST). Experiments conducted on five datasets demonstrate that the proposed model achieves SOTA performance.
- We propose a Spatio-Temporal Feature Fusion Network (STF), which integrates the Temporal Convolution Network and the Graph Attention Influence Network, adopting "Temporal-Spatial-Temporal" modeling strategy to achieve deeper spatiotemporal feature representation.
- We propose a Graph Attention Influence Network (GAIN), which dynamically updates the graph structure and selectively updates each node based on the globally most relevant node representations.
- We propose a Time Series Reconstruction Network (TSR) designed to amplify the discrepancies between original time series and their reconstructions, thereby facilitating the detection of potential anomalies. Furthermore, the Selective Feature Compression (SFC) block is embedded to reduce redundancies within the encoded representation.

2 Related work

2.1 Traditional anomaly detection

Traditional time series anomaly detection methods primarily rely on statistical theory and machine learning techniques, making them suitable for well-structured datasets with clear statistical characteristics. These methods can be broadly categorized into two types: statistical threshold-based approaches and model-driven approaches [3].

In statistical threshold-based methods, the z-score identifies anomalies by quantifying how many standard deviations a data point deviates from the dataset's mean [6]. Similarly, box plot-based methods determine anomaly thresholds using the interquartile range (IQR), which defines boundaries based on the upper and lower quartiles. These methods are simple to implement and computationally efficient, especially when applied to small datasets with clear statistical properties. However, their performance tends to degrade when handling non-linear, high-dimensional, or noisy data [7]. Moreover, their reliance on predefined thresholds or assumptions about data distribution can lead to inaccurate results, particularly in the presence of complex patterns.

Model-driven approaches, such as autoregressive (AR) [8], autoregressive moving average (ARMA) [9], and autoregres

sive integrated moving average (ARIMA) models [10], predict future values by analyzing historical data, capturing trend patterns, cyclical fluctuations, and random noise. Anomalies are detected when the actual data significantly deviates from these predictions [11]. These methods are relatively efficient and interpretable, but they are limited by their inability to handle highly non-linear relationships or datasets with irregular patterns. Additionally, they require expert knowledge for parameter tuning, which can be cumbersome when dealing with more complex datasets. Traditional methods, especially those based on statistical thresholds, are also highly sensitive to noise, often resulting in false positives when anomalies are subtle or when the data contains significant noise.

The limitations of traditional methods become particularly evident when dealing with datasets that exhibit complex temporal dependencies, high dimensionality, or non-stationary patterns [4].

2.2 Deep anomaly detection

Deep learning-based anomaly detection methods have gained significant attention in recent years due to their ability to model complex patterns and dependencies in time series. These methods can be broadly categorized into prediction-based approaches, reconstruction-based approaches, and more recently, graph-based methods, which focus on capturing relationships between variables in multivariate time series [12].

Prediction-based anomaly detection methods, such as Long Short-Term Memory (LSTM) networks and Recurrent Neural Networks (RNN), are designed to capture non-linear temporal dependencies in time series. For example, Hundman et al. [13] proposed LSTM-NDT, which utilizes LSTM to make predictions across multiple dimensions, employing a non-parametric dynamic error threshold strategy to minimize false negatives. Ergen et al. [14] extended LSTM-based models by introducing a combined strategy based on gradient and quadratic programming, optimizing the LSTM for both unsupervised and semi-supervised anomaly detection. These prediction-based methods excel in modeling long-term dependencies and capturing temporal patterns but often struggle when faced with rare or subtle anomalies. They also tend to be overly sensitive to cyclical changes [15], leading to false positives when anomalies are influenced by periodic fluctuations rather than true outliers.

Reconstruction-based anomaly detection methods, such as Variational Auto-Encoders (VAE) [16] and Generative Adversarial Networks (GAN) [17], aim to model the normal data distribution and detect anomalies based on reconstruction errors. For example, OmniAnomaly [18], which combines GRU and VAE, learns the normal patterns of multivariate time series and uses reconstruction probabilities to identify anomalies. GAN-based approaches, such as the one proposed by Audibert et al. [19], use adversarial training between a generator and discriminator to isolate anomalous data points. While these methods can successfully detect anomalies by measuring deviations in reconstruction, they often fail to capture subtle anomalies, particularly when the reconstruction error does not exceed a predefined threshold. Additionally, reconstruction-based methods may struggle to identify complex anomalies in multivariate time series due to their limited ability to capture the interactions between different variables [20].

To address the limitations of traditional deep learning methods, graph-based learning methods have emerged as a powerful tool for modeling the relationships among variables in multivariate time series. For example, Graph Deviation Networks (GDN) [21] and Graph Convolutional Networks (GCN) [22] learn to represent inter-variable relationships through graph structures. GDN uses attention mechanisms to assign importance weights to different connections, improving anomaly detection interpretability, while GCN apply graph convolutions to capture multi-scale relationships between time series variables, enhancing detection robustness. Despite their strength in modeling inter-variable relationships, graph-based methods are often limited by their inability to adequately capture temporal dependencies, which are crucial for effective time series anomaly detection.

In summary, the proposed TADST model effectively addresses these limitations. By leveraging TCN, GAIN, and AdaTSD, TADST can model both intra-variable temporal dependencies and inter-variable relationships. The TCN captures temporal dependencies within each variable, while GAIN models the intricate interactions between variables, providing a robust framework for multivariate time series anomaly detection. Additionally, the AdaTSD extracts key cyclical and trend signals, enhancing the detection of anomalies influenced by long-term temporal patterns and complex cyclical effects. This integrated approach enables TADST to outperform traditional deep learning and graph-based methods, particularly in handling noisy, high-dimensional, and non-stationary time series.

3 Preliminary

3.1 KL sparse self attention

The canonical attention mechanism faces challenges when processing long sequences, as the computational complexity grows quadratically with sequence length [23]. Additionally, the attention probability distribution typically exhibits a long-tail shape, indicating that only a small amount of key information significantly impacts performance [24].

To mitigate the interference caused by "lazy" queries in capturing long-term dependencies, as shown in Fig. 2, the

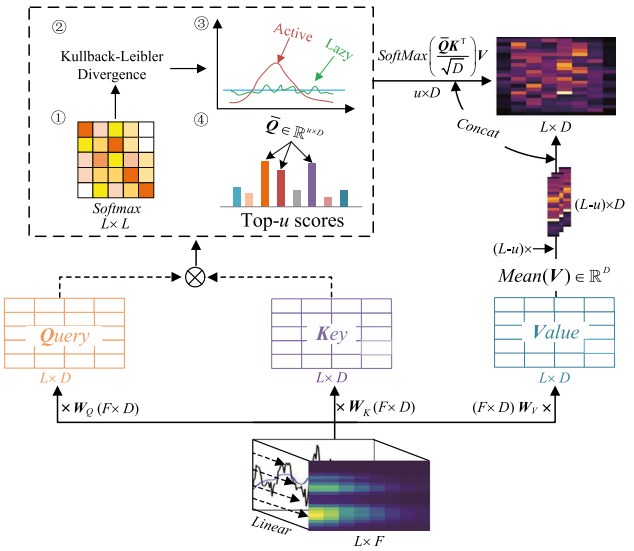


Fig. 2 Kullback-Leibler Divergence-based Sparse Self-Attention Mechanism. $\boldsymbol{Q} \in \mathbb{R}^{L\times D}$, $\boldsymbol{K} \in \mathbb{R}^{L\times D}$, and $\boldsymbol{V} \in \mathbb{R}^{L\times D}$ represent the *Query*, the *Key*, and the *Value* matrices obtained through linear mappings of the input $\boldsymbol{X} \in \mathbb{R}^{L\times F}$, where L, F, and D denote the sequence length, feature dimension, and mapping dimension, respectively

Kullback-Leibler Divergence-based Sparse Self-Attention (KL-SA) mechanism prioritizes the "active" queries. By leveraging Kullback-Leibler (KL) divergence to measure the divergence between the attention probability distribution and a uniform distribution, we can distinguish between "active" and "lazy" queries. A high KL divergence score signifies an "active" query, whereas a low score reflects a "lazy" one. The sparsity measure of a query vector $\boldsymbol{q}_i$ is defined in (1):

$$M(\boldsymbol{q}_i) = \ln \sum_{j=1}^{L} e^{\frac{\boldsymbol{q}_i \boldsymbol{k}_j^T}{\sqrt{D}}} - \frac{1}{L}\sum_{j=1}^{L} e^{\frac{\boldsymbol{q}_i \boldsymbol{k}_j^T}{\sqrt{D}}} \tag{1}$$

where the first term emphasizes significant dot products using the Log-Sum-Exp (LSE) function, highlighting the key areas of attention; the second term averages the scaled dot products, serving as a reference for uniformity. A higher $M(\boldsymbol{q}_i)$ indicates greater sparsity (focused attention on a few keys), while a lower value suggests a more even distribution of attention across keys.

Subsequently, we select top $u = \lfloor c \times \log L \rfloor$ query vectors based on sparsity scores to construct the sparse query matrix $\overline{\boldsymbol{Q}} \in \mathbb{R}^{u\times D}$, where the adjustment factor c controls the number of active queries. The KL-based Sparse Self-Attention mechanism, defined in (2), has a time complexity of $\mathcal{O}(DL \log L)$ and a space complexity of $\mathcal{O}(LD)$:

$$\text{KL-SA}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = SoftMax\left(\frac{\overline{\boldsymbol{Q}}\boldsymbol{K}^\top}{\sqrt{D}}\right)\boldsymbol{V} \oplus Mean(\boldsymbol{V}) \tag{2}$$

where $Mean(.)$ represents the averaging operation, and $\oplus$ denotes the padding of missing entries with the mean vector $L - u$ times to maintain dimensional consistency with the original attention matrix $(L \times D)$.

3.2 Temporal convolutional network

The Temporal Convolutional Network (TCN) utilizes causal and dilated convolutions, ensuring that the feature map at each timestamp is influenced only by the present and past information, thereby excluding the intervention of future data [25]. As shown in Fig. 3, by adjusting the dilation rate of convolutional kernels, the dilated convolutions can expand the receptive field.

In the multi-layer structure, each layer performs a series of causal convolutions followed by the nonlinear activation function. Dilated convolutions, controlled by the dilation rates at each layer, further enhance the ability to capture long-term dependencies:

$$\boldsymbol{Y}[t] = \sigma\left(\sum_{i=0}^{k-1} \boldsymbol{W}[i] \cdot \boldsymbol{X}[t - d * i] + b\right) + \boldsymbol{X}[t] \tag{3}$$

where k, d, and b represent the kernel size, dilation rate, and bias term, respectively, and σ denotes the ReLU activation function. This equation describes the dilated causal convolution operation of a TCN layer, with $\boldsymbol{W}[i]$ as the filter applied at time $t - d \cdot i$, and $\boldsymbol{X}[t]$ as the input at time t. The time complexity of TCN is $\mathcal{O}(LDHk)$, and its space complexity is $\mathcal{O}(LDH)$, here L, D, and H denote the sequence length, feature dimension, and number of layers, respectively.

4 Methodology

Time series anomaly detection is a binary classification task. Given a multivariate time series $\boldsymbol{X} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_L\}$, where each $\boldsymbol{x}_i \in \mathbb{R}^F$, the reconstructed sequence $\hat{\boldsymbol{X}} =$

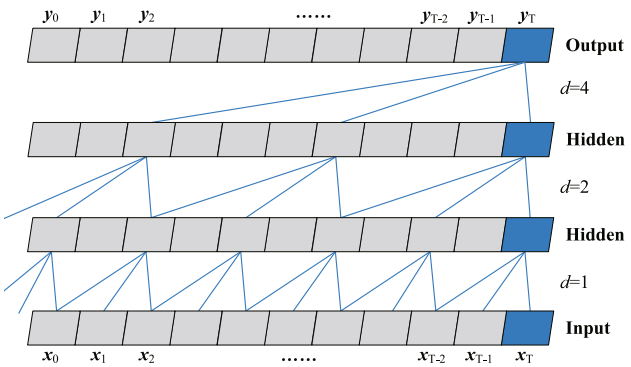


Fig. 3 Dilated Causal Convolution

$\{\hat{\boldsymbol{x}}_1, \ldots, \hat{\boldsymbol{x}}_L\}$ is obtained after processing through the Time Series Reconstruction Network. Then, the Anomaly Detection Mechanism identifies potential anomalies by measuring the deviation between $\boldsymbol{X}$ and $\hat{\boldsymbol{X}}$. Specifically, if the f-th dimensional feature value $\boldsymbol{x}_{tf}$ at timestamp t exhibits anomalous characteristics, $\boldsymbol{x}_t$ is marked as an anomaly (denoted as 1); otherwise, it is normal (denoted as 0).

4.1 Overall framework

We propose a Time Series Anomaly Detection Model Based on Spatio-Temporal Feature Fusion (TADST), as shown in Fig. 4.

First, the Spatio-Temporal Feature Fusion Network (STF) captures temporal dependencies and attribute associations to generate the spatiotemporal fusion feature (Section 4.3). Next, the Time Series Reconstruction Network (TSR), employing a encoder-decoder architecture (Section 4.4), processes the fusion feature. The encoder transforms the fusion feature into the encoded feature, while the Selective Feature Compression (SFC) reduces redundancies within the encoded feature. The decoder then analyzes the intrinsic relationships between the compressed encoded feature and the cyclical component, thereby enhancing the sensitivity of the Anomaly Detection Mechanism (ADM) by amplifying the discrepancies between reconstructed data and potential anomalies (Section 4.5).

4.2 Model input

The input $\boldsymbol{X} \in \mathbb{R}^{L \times F}$ consists of L observations $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_L\}$, where each $\boldsymbol{x}_i \in \mathbb{R}^F$ is an F-dimensional vector $\{\boldsymbol{x}_{i1}, \ldots, \boldsymbol{x}_{iF}\}$.

To mitigate the influence of sudden, non-anomalous signals during reconstruction and to ensure that TSR accounts for the contextual cyclical patterns of potential anomalies [26], the Adaptive Time Series Decomposition decomposes the original sequence $\boldsymbol{X}$ into two components: ***Trend*** and ***Cyclicality***. These components serve as inputs to the first layer of the multi-layer encoder and the decoder, respectively.

4.2.1 Adaptive time series decomposition

The traditional time series decomposition algorithm, STL, employs locally weighted regression to decompose time

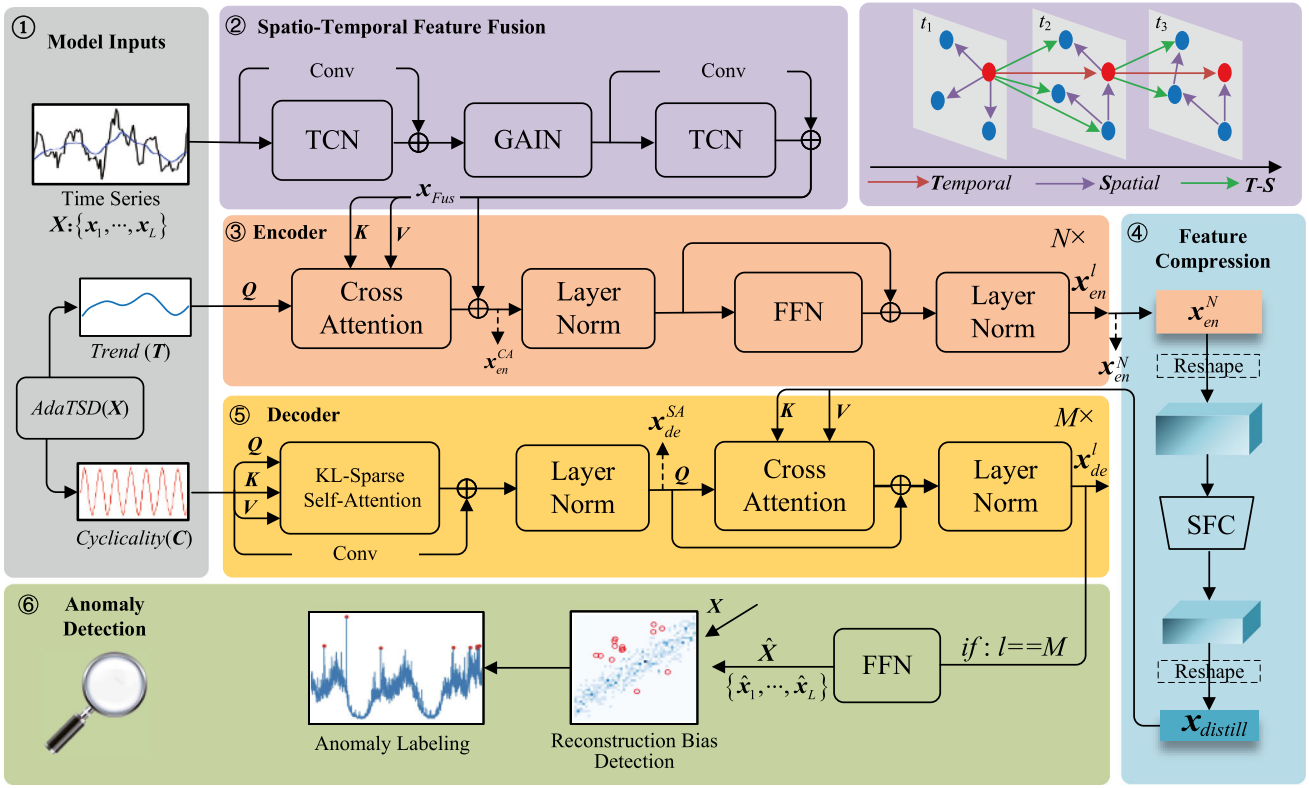


Fig. 4 Time Series Anomaly Detection Model Based on Spatio-Temporal Feature Fusion (TADST). The overall framework consists of six main modules, following the order of signal propagation. These modules generate the reconstructed sequence $\hat{\boldsymbol{X}}$, which has the same length and dimension as the input sequence $\boldsymbol{X}$. Finally, the Anomaly Detection Mechanism evaluates the degree of deviation to determine the anomaly status

series into cyclical, trend, and residual components [27]. However, the limitations of STL are its high sensitivity to hyperparameters and lack of adaptability to dynamic temporal patterns, which restricts its effectiveness in complex or dynamic settings, such as when decomposing hidden representation.

To address above challenges, we propose an Adaptive Time Series Decomposition (AdaTSD), which adaptively adjusts decomposition weights and effectiveness based on the softmax function. By alleviating issues related to hyperparameter allocation, AdaTSD significantly enhances both adaptability and efficiency of time series decomposition within neural networks. The AdaTSD method is defined in (4)-(6):

$$\boldsymbol{C}_i, \boldsymbol{T}_i = STL(\theta_i, \boldsymbol{X}\boldsymbol{W}_i), i \in \{1, \ldots, K\} \tag{4}$$

$$\beta_1, \ldots, \beta_K = SoftMax(\boldsymbol{C}_1, \ldots, \boldsymbol{C}_K) \tag{5}$$

$$\begin{cases} \boldsymbol{C} = MLP\left(\sum_{i=1}^{K} \beta_i \times \boldsymbol{C}_i\right) \\ \boldsymbol{T} = MLP\left(\sum_{i=1}^{K} \beta_i \times \boldsymbol{T}_i\right) \end{cases} \tag{6}$$

where $\boldsymbol{X}$ represents the input, θ_i denotes the specific STL parameters, and $MLP(.)$ represents a multilayer perceptron. $Softmax(.)$ computes the weights $\beta_{1\sim K}$ to merge different decomposition subcomponents into the cyclical component $\boldsymbol{C} \in \mathbb{R}^{L\times F}$ and the trend component $\boldsymbol{T} \in \mathbb{R}^{L\times F}$.

4.3 Spatio-temporal feature fusion network

Traditional time series anomaly detection methods typically analyze sequences independently in temporal domain or attribute dimension, limiting the ability to capture spatio-temporal interactions. In time series analysis, the state at each timestamp is influenced not only by its historical data but also by the effects of other features at the same moment and the past states of these features [28]. This highlights the importance of considering both temporal and spatial correlations.

To address above issues, we propose a Spatio-Temporal Feature Fusion Network (STF). First, the Temporal Convolution Network (TCN, Section 3.2) captures long-term temporal dependencies. Subsequently, the Graph Attention Influence Network (GAIN, Section 4.3.1) analyzes structural relationships among global attributes and updates the attribute representations accordingly. Finally, employing the "Temporal-Spatial-Temporal" strategy (Section 4.3.2), it progressively models and deeply integrates temporal dynamics and global attribute associations.

4.3.1 Graph attention influence network

We propose the Graph Attention Influence Network (GAIN), which combines the advantages of the attention mechanism and the graph neural network by transforming the attributes of multivariate time series into node representations within the graph structure. Then, it employs the attention mechanism to calculate spatial structural correlations among nodes, thereby updating the feature representation of each node. The time complexity of GAIN is $\mathcal{O}(LF^2)$, while the space complexity is $\mathcal{O}(LF)$.

Directed Graph Construction In the initial phase, each attribute is treated as a node $\boldsymbol{v}_i \in \mathbb{R}^L$ within the graph structure, forming the node set $\boldsymbol{V} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_F\}$, where F denotes the total number of nodes, and L denotes the feature dimension of each node.

In the absence of prior knowledge, each node $\boldsymbol{v}_i$ establishes connections with all other nodes except itself, forming a set of candidates $\boldsymbol{S}_i$. Subsequently, key dependencies are extracted from these candidates by calculating the relationship strength e_{ij} between the target node $\boldsymbol{v}_i$ and each candidate node $\boldsymbol{v}_j$, as follows:

$$e_{ij} = \frac{\boldsymbol{v}_i \boldsymbol{v}_j^T}{\|\boldsymbol{v}_i\| \cdot \|\boldsymbol{v}_j\|}, \boldsymbol{v}_j \in \boldsymbol{S}_i \tag{7}$$

where $\boldsymbol{S}_i = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_F\} \setminus \{\boldsymbol{v}_i\}$, and "\" denotes the exclusion of node $\boldsymbol{v}_i$ from the set. A higher e_{ij} value indicates a stronger dependency between nodes, allowing GAIN to effectively capture and model the spatial structural correlations inherent in multivariate time series.

To refine primary dependency relationships and connectivity patterns among nodes, the Fully Connected Undirected Graph is optimized into the Directed Graph $G(V, E)$. For each node $\boldsymbol{v}_i$, after computing the relationship strength e_{ij} with all candidate nodes $\boldsymbol{v}_j$, the Top-K strongest connected adjacent nodes are then selected:

$$A_{ij} = \{1 \mid e_{ij} \in Top_k(\{e_{ij} \mid \boldsymbol{v}_j \in \boldsymbol{S}_i\})\} \tag{8}$$

where $A_{ij} = 1$ indicates the directed edge from the adjacent node $\boldsymbol{v}_j$ to the target node $\boldsymbol{v}_i$; otherwise, $A_{ij} = 0$. Repeat this step to construct the adjacency matrix $\boldsymbol{A} \in \mathbb{R}^{F\times F}$ for the directed graph.

Edge Weight Based on the relationship strengths e_*, the directed edge weights are determined, facilitating the selective attentions of the target node towards its adjacent nodes. The attention weight α_{ij} between $\boldsymbol{v}_i$ and its adjacent node $\boldsymbol{v}_j$ is defined as:

$$\alpha_{ij} = SoftMax(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{\boldsymbol{v}_k \in N(i)} \exp(e_{ik})} \tag{9}$$

where $SoftMax(\cdot)$ transforms scores into the probability distribution, with values ranging between 0 and 1 and summing to 1; $N(i)$ represents the set of all adjacent nodes of $\boldsymbol{v}_i$.

Message Passing As shown in Fig. 5, GAIN not only propagates the representations of the node $\boldsymbol{v}_A$ and its adjacent nodes, but also considers feature differences and fusion between nodes. The state of $\boldsymbol{v}_A$ is updated based on the concatenated features and the attention weights of directed edges, as follows:

$$\boldsymbol{v}'_A = \sigma\Bigg(\sum_{\boldsymbol{v}_j \in N(A)} \boldsymbol{W}\Big(\boldsymbol{v}_A \parallel \alpha_{Aj} \cdot \boldsymbol{v}_j \parallel \alpha_{Aj} \cdot (\boldsymbol{v}_A - \boldsymbol{v}_j) \parallel \alpha_{Aj} \cdot (\boldsymbol{v}_A + \boldsymbol{v}_j) \Big) \Bigg) \tag{10}$$

where $\sigma(\cdot)$ represents the ReLU function, $\boldsymbol{v}'_A$ denotes the updated representation of the node $\boldsymbol{v}_A$, and α_* indicates the attention weight associated with a directed edge. The symbol "$\parallel$" represents the concatenation operation.

4.3.2 Feature fusion process

The "Temporal-Spatial-Temporal" feature fusion process can be formalized as:

$$\boldsymbol{x}_{Fus} = TCN(GAIN(TCN(\boldsymbol{X}))) \tag{11}$$

and is divided into three steps:

First, TCN adjusts various dilation rates to expand the receptive field, thereby capturing long-term dependencies in the temporal dimension of time series. Second, GAIN captures the interactions among attributes and updates their representations accordingly. Finally, TCN is applied again to process the hidden representation resulting from the "Temporal-Spatial" feature fusion, capturing long-term dependencies and producing the "Temporal-Spatial-Temporal" fusion representation $\boldsymbol{x}_{Fus}$. Additionally,

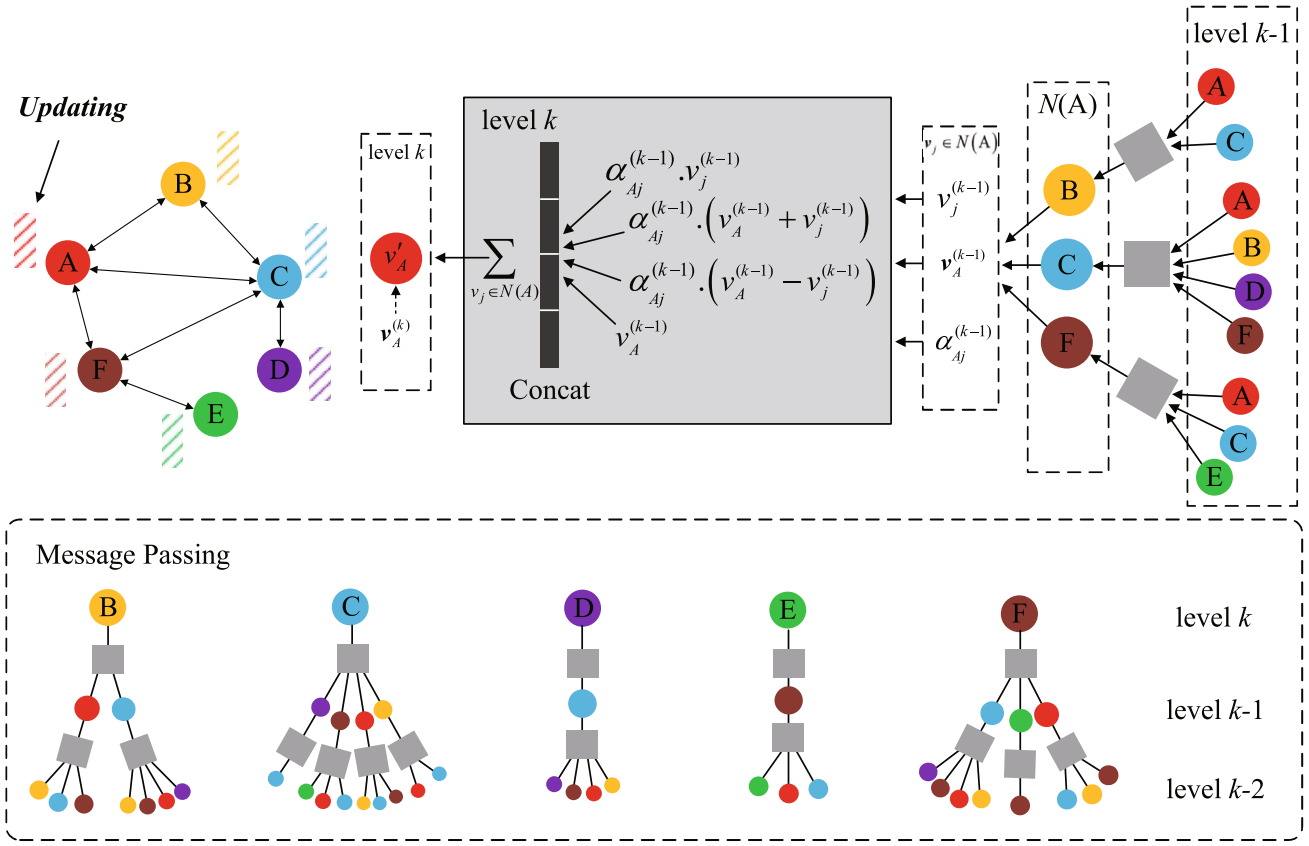


Fig. 5 Node Feature Propagation and Updating. **Top Left:** The directed graph structure that is not fully connected. **Top Right:** GAIN utilizes the set of the adjacent nodes $N(A)$ (including nodes $\boldsymbol{v}_B, \boldsymbol{v}_C, \boldsymbol{v}_F$) of the target node $\boldsymbol{v}_A$ to update the hidden representation $\boldsymbol{v}_A^{(k-1)}$ from the previous layer (k-1) based on the attention weights, thereby obtaining $\boldsymbol{v}_A$'s updated representation $\boldsymbol{v}_A^{(k)}$ at the current layer (k). **Bottom:** Each node is traversed and updated by the neural network with shared parameters. During node propagation and updating, the target node, its adjacent nodes, along with their fused features and differences, are concatenated to form the basis for the update

to address dimension mismatches in residual connections, the 1x1 convolution is employed to align the number of channels.

4.4 Time series reconstruction

The Time Series Reconstruction Network (TSR) consists of an encoder, a feature compression, and a decoder. Specifically, the encoder encodes the fusion feature (from (11)) layer by layer, while the feature compression block filters out the redundancies within the encoded feature. Based on the compressed hidden representation, the decoder reconstructs the original sequence, thereby amplifying the discrepancies between reconstructed data and potential anomalies.

4.4.1 Encoder

The encoder comprises N layers, with the l-th layer of the encoder described as

$$\boldsymbol{x}_{en}^{l} = Encoder(\boldsymbol{x}_{en}^{l-1}, \boldsymbol{x}_{Fus}) \tag{12}$$

where $l \in \{1,\ldots,N\}$. This recursive structure establishes the multi-layer encoder, yielding the final encoded feature, denoted as $\boldsymbol{x}_{en}^{N}$. The encoder operation is as follows:

In the first encoding layer, as defined in (13)-(14), the cross attention (CA) mechanism employs the ***Trend*** component as the ***Query***, while the temporal-spatio-temporal fusion feature $\boldsymbol{x}_{Fus}$ simultaneously serves as both the ***Key*** and the ***Value*** [29]. The trend component primarily captures long-term trend patterns, whereas the fusion feature provides detailed contextual spatiotemporal information.

$$\begin{cases} \boldsymbol{Q} = \boldsymbol{W}_{Q}\left(\boldsymbol{T}\cdot\delta_{l,1} + \boldsymbol{x}_{en}^{l-1}\cdot(1-\delta_{l,1})\right) \\ \boldsymbol{K} = \boldsymbol{W}_{K}\boldsymbol{x}_{Fus} \\ \boldsymbol{V} = \boldsymbol{W}_{V}\boldsymbol{x}_{Fus} \end{cases} \tag{13}$$

$$\boldsymbol{x}_{en}^{CA} = CA(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) + \boldsymbol{x}_{Fus} \tag{14}$$

where $\delta_{l,1}$ denotes the Kronecker Delta function, which is 1 when $l = 1$ and 0 otherwise.

In subsequent layers of the encoder, the cross-attention mechanism takes the encoded feature $\boldsymbol{x}_{en}^{l-1}$ from the previous layer (l-1) as the new ***Query***. This process repeats, forming the multi-layer encoder structure.

As the depth of the neural network increases, updates to parameters in each layer affect the input distribution of subsequent layers. This phenomenon, known as Internal Covariate Shift (ICS), requires each layer to continuously adapt to variations in the output distribution of its preceding layers, thereby increasing the training complexity. To address this challenge, the implementation of the stacked normalization framework (15) effectively minimizes inter-layer input distribution shifts, enhancing the stability and convergence of training process.

$$\boldsymbol{x}_{en}^{l} = Norm(FFN(Norm(\boldsymbol{x}_{en}^{CA}))) \tag{15}$$

where $Norm(.)$ denotes the LayerNorm layer, and $FFN(.)$ denotes the Feedforward Neural Network.

4.4.2 Feature compression

The multi-layer encoder may capture noise and irrelevant details within the hidden representation, leading to overfitting of the model. To mitigate this issue, we embed the Selective Feature Compression (SFC) block between the encoder and the decoder, which aims to (***i***) eliminate redundancies and noise while preserving essential semantic information, and (***ii***) reduce the feature dimension to improve computational efficiency.

Selective Feature Compression Inspired by the self-attention mechanism [30], the Selective Feature Compression (SFC) focuses on extracting key information and compressing redundant information (Fig. 6). The resulting hidden representation $\boldsymbol{x}_{distill}$ exhibits the following characteristics: (i) the unchanged sequence length L, and (ii) the compressed feature dimension $\frac{D}{\tau}$. The time complexity of SFC is $\mathcal{O}(LD\cdot\frac{D}{\tau})$, and its space complexity is $\mathcal{O}(L\cdot\frac{D}{\tau})$, where τ is the downsampling rate.

To accomplish this, we first add an additional channel dimension to the 2-D encoded feature $\boldsymbol{x}_{en}^{N} \in \mathbb{R}^{L\times D}$, transforming it into the 3-D form $\boldsymbol{F}_{head} \in \mathbb{R}^{L\times\frac{D}{C}\times C}$. Subsequently, the most relevant features are selected along the channel dimension with downsampling rate τ to achieve compression. Finally, the 3-D representation is reshaped into the 2-D representation $\boldsymbol{x}_{distill} \in \mathbb{R}^{L\times\frac{D}{\tau}}$. The SFC method is as follows:

$$\begin{aligned} \boldsymbol{x}_{distill} &= SFC(\boldsymbol{Q}=\boldsymbol{K}:\boldsymbol{F}_{abs};\boldsymbol{V}:\boldsymbol{F}_{head}) \\ &= Softmax(\theta_{Q}(pool(\boldsymbol{Q}))^{T}\cdot\theta_{K}(\boldsymbol{K}))\cdot\boldsymbol{V} \end{aligned} \tag{16}$$

where θ_{*} represents the linear layer implemented by a 2D convolution kernel; $pool(.)$ is the Top-K pooling layer; $\boldsymbol{F}_{abs} = \Theta_{abs}(\boldsymbol{F}_{head}) \in \mathbb{R}^{L\times\frac{D}{2C}\times C}$ is the re-encoding operation for the head feature $\boldsymbol{F}_{head}$, reducing the feature dimension by half through the Residual Network (ResNet). Note that for simplicity, the reshaping function $R(.)$ is omitted in (16) but included in Fig. 6.

Top-K Pooling The Top-K pooling layer downsamples the ***Query***, reducing the channels from C to $\frac{C}{\tau}$. Unlike the mean or max pooling [31], the Top-K pooling does not restrict itself to compressing local features within small windows. Instead, it pools $\boldsymbol{F}_{abs} \in \mathbb{R}^{L\times\frac{D}{2C}\times C}$ with the downsampling rate τ, resulting in $\boldsymbol{F}_{pool} \in \mathbb{R}^{L\times\frac{D}{2C}\times\frac{C}{\tau}}$. This process selectively

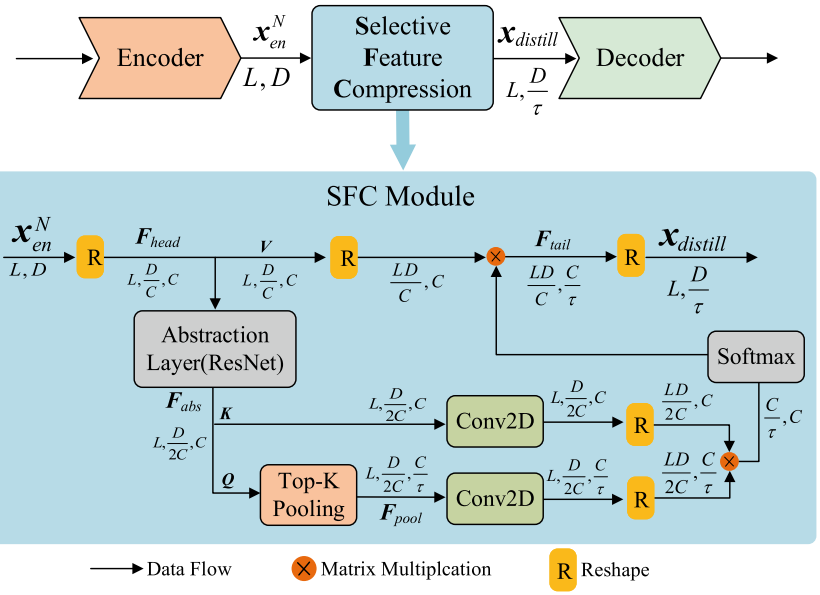


Fig. 6 Selective Feature Compression

retains the top-priority $\frac{C}{\tau}$ channels, effectively compressing features while preserving key information.

4.4.3 Decoder

The decoder comprises M layers, with the l-th layer of the decoder described as

$$\boldsymbol{x}_{de}^{l} = Decoder(\boldsymbol{x}_{de}^{l-1}, \boldsymbol{x}_{distill}) \tag{17}$$

where $l \in \{1, \ldots, M\}$. This recursive structure establishes the multi-layer decoder, yielding the final decoded feature, denoted as $\boldsymbol{x}_{de}^{M}$. The decoder operation is as follows:

In the first decoding layer, the KL Sparse Self-Attention (KL-SA) mechanism (Section 3.1) processes $\boldsymbol{C}yclical$ information, more accurately distinguishing true anomalies from normal fluctuations and effectively reducing the risk of mistakenly classifying normal behavior as anomalous. Subsequently, the layer normalization is applied to the outputs of both the self-attention and the cross-attention (CA) mechanisms (19).

In subsequent layers of the decoder, KL-SA takes the decoded feature $\boldsymbol{x}_{de}^{l-1}$ from the previous layer (l-1) as its input. The compressed encoded feature $\boldsymbol{x}_{distill}$ serves as both the ***Key*** and the ***Value*** for the cross-attention mechanism, while $\boldsymbol{x}_{de}^{SA}$ serves as the ***Query***. This process repeats, forming the multi-layer decoder structure:

$$\boldsymbol{x}_{de}^{l-1} = \boldsymbol{C} \cdot \delta_{l,1} + \boldsymbol{x}_{de}^{l-1} \cdot (1 - \delta_{l,1}) \tag{18}$$

$$\begin{cases} \boldsymbol{x}_{de}^{SA} = Norm(KL\text{-}SA(\boldsymbol{x}_{de}^{l-1}) + \boldsymbol{x}_{de}^{l-1}) \\ \boldsymbol{x}_{de}^{l} = Norm(CA(\boldsymbol{x}_{de}^{SA}, \boldsymbol{x}_{distill}) + \boldsymbol{x}_{de}^{SA}) \end{cases} \tag{19}$$

where $\delta_{l,1}$ denotes the Kronecker Delta function, $Norm(.)$ represents the layer normalization, $CA(.)$ denotes the cross-attention mechanism, and $\boldsymbol{C}$ denotes the cyclical component decomposed by the adaptive time series decomposition (6).

After decoding through successive layers, the FeedForward Neural Network (FFN) maps the decoded feature $\boldsymbol{x}_{\mathrm{de}}^{M}$ into F-dimensional space, ensuring that the reconstructed data point $\hat{\boldsymbol{x}}_i \in \mathbb{R}^F$ matches the dimension of the original data point $\boldsymbol{x}_i \in \mathbb{R}^F$:

$$\begin{aligned} \hat{\boldsymbol{x}}_1, \hat{\boldsymbol{x}}_2, \ldots, \hat{\boldsymbol{x}}_L &= FFN(\boldsymbol{x}_{de}^{M}) \\ &= \sigma(\boldsymbol{x}_{de}^{M}\boldsymbol{W}_1 + \boldsymbol{b}_1)\boldsymbol{W}_2 + \boldsymbol{b}_2 \end{aligned} \tag{20}$$

where the activation function σ is the Leaky ReLU function.

4.5 Anomaly detection mechanism

In multivariate time series, outliers or anomalies may manifest in specific dimensions while others remain unaffected. To address this, the Anomaly Detection Mechanism (ADM) considers the overall deviation in the reconstructed multivariate data as follows:

For the predicted value $\hat{\boldsymbol{x}}_t$ and its corresponding observed value $\boldsymbol{x}_t$ at timestamp t, the reconstruction deviation is defined as:

$$e_t = \max_{f=1}^{F} |\boldsymbol{x}_{tf} - \hat{\boldsymbol{x}}_{tf}| \tag{21}$$

where the deviation series is $\boldsymbol{e} = \{e_1, \ldots, e_L\}$; L and F denote the sequence length and the number of attributes, respectively.

Given that abrupt changes in time series often lead to peaks in prediction errors, resulting in a higher false alarm rate in

anomaly detection, the Exponential Weighted Moving Average (EWMA) is applied to smooth the deviation series $\boldsymbol{e}$. The anomaly score s_t is defined as:

$$s_t = s_{t-1} \cdot \beta + e_t \cdot (1 - \beta) \tag{22}$$

where s_t and e_t represent the anomaly score and the reconstruction deviation, respectively; the smoothing coefficient β adjusts the influence of historical anomaly scores on the current anomaly score.

Subsequently, the Generalized Pareto Distribution (GPD) is fitted to the tail distribution of anomaly scores $\{s_t\}$. By performing maximum likelihood estimation of the parameters, this method characterizes the statistical properties of extreme events. The probability density function $f(s; \xi, \mu, \sigma)$ is defined as:

$$f(s;\xi,\mu,\sigma) = \begin{cases} \frac{1}{\sigma}\left(1+\xi\frac{s-\mu}{\sigma}\right)^{-\left(1+\frac{1}{\xi}\right)}, & \xi \neq 0 \\ \frac{1}{\sigma}\exp\left(-\frac{s-\mu}{\sigma}\right), & \xi = 0 \end{cases} \tag{23}$$

where σ is the scale parameter, ξ is the shape parameter, and μ is the location parameter representing the current threshold. The tail distribution fitting is applicable for $s \geq \mu$.

During continuous monitoring, each new data point is evaluated against the latest anomaly threshold. If the anomaly score of a data point exceeds this threshold, it is classified as an anomaly. Such an event triggers the re-fitting and updating of the GPD parameters, refining the detection criteria for subsequent data points. The updated threshold is calculated as:

$$\tau_{new} = \tau_{prev} + \frac{\sigma}{\xi}\left[\left(\frac{N_\tau}{N \times \alpha}\right)^{\xi} - 1\right] \tag{24}$$

where σ and ξ are the scale and shape parameters; N_τ denotes the number of samples currently identified as anomalies, N represents the total number of samples, and α is the expected proportion of anomalies. When this ratio $\frac{N_\tau}{N \times \alpha}$ exceeds 1, it indicates that more anomalies than expected have been detected, suggesting the threshold should be raised to mitigate false positives. Conversely, when the ratio is less than 1, it implies fewer anomalies than expected, indicating that the threshold should be lowered to capture more anomalies and reduce false negatives.

4.6 Training strategy

The proposed TADST model is composed of three main components: the Spatio-Temporal Feature Fusion Network (STF, Section 4.3), the Time Series Reconstruction Network (TSR, Section 4.4), and the Anomaly Detection Mechanism (ADM, Section 4.5). To effectively train the model, the phased training strategy is employed, consisting of two stages:

Reconstruction Network Training During this phase, STF and TSR are jointly optimized, while ADM remains frozen. The Mean Squared Error (MSE) is employed as the loss function:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{x}_i - \hat{\boldsymbol{x}}_i)^2 \tag{25}$$

where $\boldsymbol{x}_i$ and $\hat{\boldsymbol{x}}_i$ denotes the original and the reconstructed data point, respectively.

Classification Network Training Once the reconstruction network is pre-trained, the second phase begins by unfreezing ADM. During this phase, TADST is treated as an integrated model, with the Binary Cross-Entropy Loss (BCE) employed as the loss function:

$$BCE = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right] \tag{26}$$

where p_i represents the probability; y_i indicates the status, with anomalies labeled as 1 and normal instances as 0.

5 Experiments

5.1 Datasets

The experiments evaluate the performance of models across five publicly available datasets. For dataset preparation, all datasets follow a consistent structure, with the training and test sets clearly defined by the official sources.

In our experiments, results were evaluated based on the test set, and the training set was split into two subsets: 70% for training and 30% for validation. Statistical information is presented in Table 1, as detailed below.

- **PSM** [32] contains monitoring data from multiple servers at eBay, consisting of 25 features that detail server

Table 1 Dataset Statistical Information

Dataset	Features	Train	Test	Anomaly Ratio (%)
PSM	25	132,481	87,841	27.8
SWaT	51	475,200	449,919	12.2
DSADS	45	85,500	57,000	9.6
SWAN	9	60,000	60,000	32.6
GECCO	38	69,260	69,261	1.1

performance metrics. It presents challenges such as diverse anomaly patterns and high complexity, with anomalies like sudden fluctuations, system crashes, and performance degradation.

- **DSADS** [12] includes sensor data for 19 different daily life and sports activities, such as accelerometers, gyroscopes, and magnetometers. To simulate normal and anomalous data, activities with higher intensity and lower frequency are categorized as anomalies, while the rest are considered normal. The main challenges are numerous activity types with low distinguishability and sample imbalance.
- **SWaT** [33] originates from a water treatment platform, collected by 51 continuously operating sensors over a period of 11 days, including 7 days of normal operations and 4 days of anomalous data. It faces challenges such as class imbalance, complex anomaly patterns, and the curse of dimensionality due to a large number of features.
- **NIPS-TS-SWAN** [34] comes from NASA's global monitoring program for solar and space weather, providing high-quality data on solar surface activity areas. The challenges include diverse anomaly patterns and data fluctuations, complicating anomaly detection in the presence of fluctuating solar activity.
- **NIPS-TS-GECCO** [35] stems from the 2018 Genetic and Evolutionary Computation Conference, focusing on the detection of drinking water quality in IoT environments. Its challenges include class imbalance and missing data issues due to sensor malfunctions and environmental factors.

5.2 Data preprocessing

To ensure the quality and reliability of the experimental results, several preprocessing and data augmentation techniques were applied to address issues such as data standardization, missing values, and class imbalance.

5.2.1 Handling missing values

The datasets used in the experiments contained some data points with missing values (NA) or infinite values (INF). These values were replaced with zeros using a zero imputation strategy to maintain dataset integrity. The zero-filled entries were labeled as anomalous (label = 1), as missing values often occur due to issues during data collection or measurement and, in many cases, represent a form of anomaly in the data.

5.2.2 Data normalization

Z-score normalization was applied instead of Min-Max normalization. Z-score normalization is less sensitive to outliers, which is crucial for time series anomaly detection. In contrast, Min-Max normalization can be influenced by extreme values, distorting the distribution of features [36].

5.2.3 Mitigating class imbalance

To address the issue of class imbalance, particularly the scarcity of anomalous samples, we employed the Synthetic Minority Over-sampling Technique (SMOTE). The process included:

Step1. Selection of Normal Samples: Normal samples were randomly selected from the training set.

Step2. Generation of Synthetic Samples: SMOTE interpolated between the selected normal samples and their nearest neighbors to generate synthetic samples.

Step3. Adding Gaussian Noise: Gaussian noise was added to the synthetic samples. High-noise samples were labeled as anomalous (label = 1), while low-noise samples were labeled as normal (label = 0).

This approach increased the number of anomalous samples and simulated diverse anomalies, thereby enhancing the model's ability to detect complex abnormal patterns.

5.3 Implementation details

TADST was trained for 600 epochs with a batch size of 32 samples. For parameter optimization, the Adam optimizer was utilized with the learning rate of 0.0001. Additionally, the sequence length L was set to 50 in Section 5.6. To mitigate overfitting, an early stopping strategy was implemented to monitor the efficiency and effectiveness of the training process. The parameters of TADST are presented in Table 2.

5.4 Baseline models

We select 13 models for benchmarking, described as follows:

LOF [37] detects outliers by comparing the density of an object in its local neighborhood with that of its neighbors. OCSVM [38] learns the decision boundary for normal observations, and values outside this boundary are deemed anomalies. AutoEncoder [39] uses the encoder to compress feature into lower-dimensional space and the decoder to reconstruct it, minimizing reconstruction error. DeepSVDD [40] maps normal samples into high-dimensional feature space and constructs a hypersphere. DAGMM [41] combines the deep autoencoder with the gaussian mixture model, where the autoencoder learns low-dimensional representation, and the gaussian mixture model estimates the density. OmniAnomaly [18] integrates gated recurrent units and variational autoencoders to learn the normal patterns of time series. MTAD-GAT [42] analyzes temporal relations using two parallel graph attention network layers. NeuTraL [43] embeds transformed data into semantic space while preserving the

Table 2 TADST parameters

Inputs Module:	
AdaTSD	*s.window*:{2,4,8,16} *robust*:True *t.window*:{2,4,8,16}
STF:	
TCN	*kernelSize*:3 LeakyReLU(.)
GAIN	*layer*:3 *Head*:4
Encoder:	*N*:2
Cross Attention	*dmodel*:1024 *Head*:8
	Skip-Connect LayerNorm(.)
FFN	Conv1D(*Inchannels*:1024, *Outchannels*:2048)
	Conv1D(*Inchannels*:2048, *Outchannels*:1024)
Decoder:	*M*:2
KL-SA	*dmodel*:1024 *Head*:8 *c*:10
	Dropout(0.1) LayerNorm(.)
Cross Attention	*dmodel*: 1024 *Head*:8
	LayerNorm(.) Skip-Connect
FFN	Conv1D(*Inchannels*:1024, *Outchannels*:512)
	Conv1D(*Inchannels*:512, *Outchannels*:*F*)
ADM:	
EWMA	β:0.1

original similarity. ECOD [44] initially estimates variable distributions in a non-parametric way, then multiplies the estimated tail probabilities across all dimensions to obtain an anomaly score for observations.

The following represent the latest models: COUTA [45] combines uncertainty and local anomaly calibration strategies to enhance the credibility of predictions for normal samples. Dual-TF [15] uses nested sliding windows in both temporal and frequency domains to address granularity, allowing better alignment of anomaly scores. NPSR [46] proposes a "normality score" metric based on combined reconstruction errors, deriving an induced anomaly score by integrating with existing anomaly scores. TADNet [47] employs STL to attribute different types of anomalies to specific components of the decomposition, simplifying the analysis of complex time series.

5.5 Evaluation metrics

We utilize Precision (P), Recall (R), and F1 score (F1) as evaluation metrics to comprehensively assess the performance of models in detecting anomalies. The F1-score is defined as follows:

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{27}$$

5.6 Comparison experiments

5.6.1 Results on the PSM, SwaT, and DSADS

In this section, the proposed TADST model is compared with the 13 baseline models on the PSM, SwaT, and DSADS datasets. As shown in Table 3, TADST achieves SOTA performance in terms of F1 scores. Specifically, compared to AutoEncoder, TADST demonstrates improvements of 14.02%, 80.31%, and 21.69% in F1 scores across the three respective datasets.

Traditional anomaly detection algorithms, such as LOF and OCSVM, perform well on specific metrics in certain scenarios. However, their overall performance is generally limited due to the inability to effectively leverage temporal information and their constraints in handling high-dimensional sequences. For example, while OCSVM achieves high recall (0.899) on the PSM dataset, its precision is only 0.682, resulting in relatively low F1 score (0.775).

Autoencoder-based methods, such as AutoEncoder, DAGMM, and OmniAnomaly, address the challenge of handling high-dimensional hidden representation. However, these approaches do not achieve an optimal balance between precision and recall. This shortcoming hinders effective differentiation between anomalous and normal instances, especially in scenarios with frequent non-anomalous fluctuations. For example, on the SwaT dataset, while AutoEncoder achieves the recall of 0.949, its precision is only 0.352, resulting in some normal instances being misclassified as anomalies.

In contrast, TADST achieves a superior balance between precision and recall. This is due to its comprehensive consideration of potential dynamic patterns in time series, temporal dependencies, and contextual information. It leverages the advantages of the Graph Attention Influence Network for processing structured data and the target-oriented information focusing capability of the KL Sparse Attention mechanism [23]. Moreover, the redundancy filtering technology also enhances overall performance [48]. This balanced performance demonstrates TADST's ability to reduce false positives while minimizing missed detections of true anomalies.

In summary, the proposed TADST demonstrates exceptional anomaly detection capabilities across the PSM, SwaT, and DSADS datasets, showcasing its wide applicability in the field of time series anomaly detection.

Table 3 Quantitative Results on the PSM, SwaT, and DSADS Datasets

Model	DataSet								
	PSM			SwaT			DSADS		
	P	R	F1	P	R	F1	P	R	F1
LOF	0.668	0.892	0.763	**0.995**	0.731	0.842	0.467	0.918	0.619
OCSVM	0.682	0.899	0.775	0.975	0.736	0.838	0.745	0.942	0.832
AutoEncoder	**1.000**	0.705	0.827	0.352	0.949	0.513	0.627	**1.000**	0.770
DeepSVDD	0.881	0.850	0.865	0.963	0.840	0.897	0.750	0.849	0.796
DAGMM	0.721	**0.989**	0.834	0.356	**0.979**	0.522	0.511	0.748	0.607
OmniAnomaly	0.901	0.923	0.911	0.781	0.740	0.759	0.859	0.885	0.871
MTAD-GAT	0.776	0.899	0.833	0.987	0.723	0.834	0.812	0.960	0.879
NeuTraL	0.937	0.919	0.927	0.993	0.756	0.858	0.829	0.906	0.865
ECOD	0.967	0.863	0.912	0.933	0.777	0.847	0.876	0.977	0.923
COUTA*	0.985	0.840	0.906	0.908	0.867	0.887	0.897	0.966	0.930
Dual-TF*	0.965	0.892	0.927	0.897	0.936	0.916	0.894	0.965	0.928
NPSR*	0.991	0.896	0.941	0.921	0.886	0.903	0.903	0.969	0.935
TADNet*	0.979	0.893	0.934	0.905	0.938	0.921	**0.912**	0.955	0.933
TADST(Ours)	0.982	0.907	**0.943**	0.919	0.931	**0.925**	0.895	0.983	**0.937**

Note: The best results are highlighted in **bold**, and the second-best results are underlined. Models marked with * represent the latest SOTA

5.6.2 Results on the NIPS-TS

The NIPS-TS-GECCO and NIPS-TS-SWAN datasets present challenges such as contextual anomalies and collective anomalies, which contribute to the increased complexity of anomaly detection tasks.

As shown in Table 4, the proposed TADST is evaluated on the GECCO and SWAN datasets. Compared to OmniAnomaly, TADST achieves F1 score improvement of 21.58% on the GECCO dataset and 4.21% on the SWAN dataset. Furthermore, when compared to the second-best model, TADST demonstrates performance improvements of 2.94% and 0.92% on the GECCO and SWAN datasets, respectively.

Table 4 Quantitative Results on the NIPS-TS Dataset

Model	DataSet					
	NIPS-TS-GECCO			NIPS-TS-SWAN		
	P	R	F1	P	R	F1
LOF	0.083	0.349	0.134	0.517	**0.794**	0.626
OCSVM	0.954	0.346	0.508	0.456	0.603	0.519
AutoEncoder	0.032	**1.000**	0.062	0.670	0.763	0.713
DeepSVDD	0.496	0.256	0.338	0.999	0.585	0.738
DAGMM	0.835	0.320	0.462	0.938	0.635	0.757
OmniAnomaly	**0.998**	0.351	0.519	0.999	0.585	0.737
MTAD-GAT	0.966	0.354	0.518	**0.999**	0.607	0.755
NeuTraL	0.988	0.352	0.519	0.537	0.784	0.637
ECOD	0.906	0.383	0.538	0.910	0.650	0.758
COUTA*	0.969	0.347	0.511	0.899	0.649	0.753
Dual-TF*	0.954	0.428	0.591	0.879	0.637	0.739
NPSR*	0.912	0.462	0.613	0.903	0.658	0.761
TADNet*	0.835	0.457	0.591	0.895	0.650	0.753
TADST(Ours)	0.809	0.517	**0.631**	0.904	0.667	**0.768**

Note: The best results are highlighted in **bold**, and the second-best results are underlined. Models marked with * represent the latest SOTA

5.7 Ablation study

To assess the contributions of each component of TADST, ablation experiments are conducted on the PSM, SwaT, and DSADS datasets as shown in Table 5, specifically:

(w/o)KL-SA This variant removes the Kullback-Leibler Divergence-based Sparse Self-Attention (KL-SA), and instead utilizes canonical attention [23] to capture long-range dependencies. Due to the long-tail characteristics commonly observed in attention distribution, removing KL-SA leads to increased redundant attention.

(w/o)AdaTSD This variant removes the Adaptive Time Series Decomposition (AdaTSD), and resorts to the STL algorithm [27]. Compared to AdaTSD, the traditional time series decomposition algorithm is limited in its synergy with neural networks. Specifically, the F1 scores drop by 0.85%, 0.97%, and 1.17% on the three datasets, respectively.

(w/o)GAIN This variant removes the Graph Attention Influence Network (GAIN), and instead utilizes CNN to capture local features. Compared to CNN [49], GAIN better perceives global attribute association and updates representations based on the attention weights. Specifically, the F1 scores drop by 1.06%, 1.51%, and 0.85% on the three datasets, respectively.

(w/o)TCN This variant removes the Temporal Convolutional Network (TCN), and instead utilizes LSTM to capture temporal dependencies. Compared to TCN, LSTM faces risks of gradient vanishing or exploding when dealing with long sequences [14].

(w/o)SFC After removing the Selective Feature Compression (SFC), the encoded feature contains more redundant hidden representations, leading to decreased decoding efficiency. Specifically, the F1 scores drop by 1.48%, 1.30%, and 1.71% on the three datasets, respectively.

(w/o)[AdaTSD+KL-SA] The simultaneous removal of AdaTSD and KL-SA leads to a significant performance drop. The absence of these components weakens the ability to perform temporal feature decomposition and model long-term dependencies. On the PSM, SwaT, and DSADS datasets, the F1 scores decrease by 1.28%, 0.54%, and 0.21%, respectively, indicating that the temporal decomposition and sparse attention mechanism are critical for TADST.

(w/o)[TCN+GAIN] The simultaneous removal of TCN and GAIN results in the model losing its ability to efficiently extract temporal information and build global attribute representations. This leads to F1 score decreases of 1.95%, 2.54%, and 2.07% on the PSM, SwaT, and DSADS datasets, respectively. These results demonstrate the essential role of TCN and GAIN in modeling spatio-temporal features.

(w/o)[TCN+GAIN+SFC] The simultaneous removal of TCN, GAIN, and SFC causes further performance degradation. The lack of SFC introduces significant redundancy during encoding, reducing decoding efficiency. On the PSM, SwaT, and DSADS datasets, the F1 scores drop by 2.27%, 2.21%, and 2.74%, respectively, emphasizing the importance of the synergistic effects of these modules in improving model performance.

(w/o)[KL-SA+TCN+GAIN+SFC] The removal of KL-SA, TCN, GAIN, and SFC together severely weakens the model's ability to capture temporal relationships, model global attribute dependencies, and perform feature compression. On the PSM, SwaT, and DSADS datasets, F1 scores decrease by 2.83%, 4.05%, and 3.88%, respectively, representing the most significant performance drop across all ablation combinations. These results underscore the indispensable role of these modules in the overall model architecture.

The ablation study results demonstrate the critical contribution of each component in the TADST to overall performance. Among them, the AdaTSD, the GAIN, and the SFC play particularly vital roles. The synergistic interaction between these modules significantly enhances the model's

Table 5 Ablation Study Results

Version	DataSet								
	PSM			SwaT			DSADS		
	P	R	F1	P	R	F1	P	R	F1
TADST	0.982	0.907	0.943	0.919	0.931	0.925	0.895	0.983	0.937
(w/o)KL-SA	0.979	0.906	0.941	0.908	0.947	0.927	0.887	0.984	0.933
(w/o)AdaTSD	0.972	0.902	0.935	0.915	0.917	0.916	0.902	0.951	0.926
(w/o)GAIN	0.975	0.894	0.933	0.923	0.899	0.911	0.912	0.947	0.929
(w/o)TCN	0.984	0.912	0.947	0.911	0.923	0.917	0.898	0.975	0.935
(w/o)SFC	0.971	0.890	0.929	0.921	0.905	0.913	0.883	0.961	0.921
(w/o)[AdaTSD+KL-SA]	0.974	0.892	0.931	0.914	0.926	0.920	0.889	0.987	0.935
(w/o)[TCN+GAIN]	0.968	0.886	0.925	0.917	0.887	0.902	0.876	0.963	0.918
(w/o)[TCN+GAIN+SFC]	0.970	0.879	0.922	0.921	0.891	0.905	0.869	0.959	0.912
(w/o)[KL-SA+TCN+GAIN+SFC]	0.988	0.856	0.917	0.868	0.912	0.889	0.864	0.946	0.902

ability to detect anomalies across multiple datasets, further validating the rationale and necessity of the proposed architecture.

5.8 Sensitivity analysis

5.8.1 Window size sensitivity

We evaluate the impact of window size $L \in \{5, 10, 25, 50, 75, 100, 125, 150\}$ on the performance of TADST.

As shown in Table 6 and Fig. 7, taking the PSM dataset as an example, when the window size is 5, TADST struggles to capture the characteristics of anomalous data due to the lack of sufficient contextual information, resulting in lower recall (0.847) and missing many true anomalies. As the window size increases to 100, the Temporal Convolutional Network and Sparse Self-Attention Mechanism are able to capture long-range dependencies and attribute associations. Compared to a window size of 5, the F1 score improves by 6.58%.

When the window size exceeds 100, TADST's efficiency in capturing temporal dependencies declines, leading to performance degradation. This phenomenon is also validated across the SwaT and DSADS datasets. However, compared to the models in Table 3, TADST outperforms most models under extreme conditions compared to their performance under normal conditions. This advantage arises from TADST's architecture, where, for example, the Adaptive Time Series Decomposition extracts trend and cyclical information.

Specifically, window sizes struggle to capture long-term patterns [27], but by introducing a trend component, TADST mitigates potential "local overfitting" and smooths short-term fluctuations, reducing noise interference during anomaly detection [47]. Additionally, cyclicality helps distinguish regular fluctuations from true anomalies [23, 26], enhancing robustness. As shown in Table 6, TADST remains stable even with window sizes smaller than 75 or larger than 125.

In summary, TADST effectively accounts for the impact of window size variation. Given the differing characteristics of time series, we determine the optimal window size or range through experimentation. As shown in Table 6, TADST does not rely on a single window size but performs better within a range of window sizes (e.g., 75-125). This demonstrates that our model exhibits strong robustness.

5.8.2 Parameter sensitivity

We evaluate the impact of various parameters on performance across the PSM, SwaT, and DSADS datasets, as shown in Fig. 8. The optimal configurations were determined based on experimental results:

Number of Fusion Layers As shown in Fig. 8(a), the performance of TADST increases with more layers in the Spatio-Temporal Fusion Network, as additional layers enable the model to capture deeper spatio-temporal correlations. However, beyond two layers, increased model complexity leads to overfitting and reduced sensitivity to anomalies. A grid search over 1-5 layers was conducted, evaluating F1-scores on a validation set. While the optimal number of layers varies across datasets, two layers provide a balanced trade-off, delivering consistent and robust performance, as highlighted by the grey dashed line.

Encoder and Decoder Layers Figure 8(b) and (c) illustrate the impact of varying the number of layers in the encoder and decoder, respectively. With fewer layers, the model fails to effectively encode and decode feature representations. Conversely, adding more than two layers yields diminishing performance gains and may introduce overfitting. Consequently, two layers were chosen for both the encoder and decoder, offering the best balance between performance and model complexity.

Number of Attention Heads Figure 8(d) shows the impact of varying attention heads on model performance. Increasing the number of heads improves the model's ability to capture diverse dependencies and relationships within the input sequence, as each head focuses on distinct attention patterns.

Table 6 Quantitative Results of Different Window Sizes in Anomaly Detection

Window Size	DataSet								
	PSM			SwaT			DSADS		
	P	R	F1	P	R	F1	P	R	F1
5	0.951	0.847	0.896	0.887	0.915	0.901	0.883	0.924	0.903
10	0.967	0.885	0.924	0.915	0.919	0.917	0.889	0.964	0.925
25	0.975	0.917	0.945	0.918	0.928	0.923	0.901	0.942	0.921
50	0.982	0.907	0.943	0.919	0.931	0.925	0.895	**0.983**	0.937
75	**0.989**	0.916	0.951	0.922	0.940	**0.931**	0.901	0.980	0.939
100	0.984	**0.928**	**0.955**	**0.926**	0.928	0.927	0.916	0.982	**0.948**
125	0.969	0.901	0.934	0.901	**0.946**	0.923	**0.921**	0.962	0.941
150	0.957	0.880	0.917	0.884	0.938	0.910	0.908	0.941	0.924

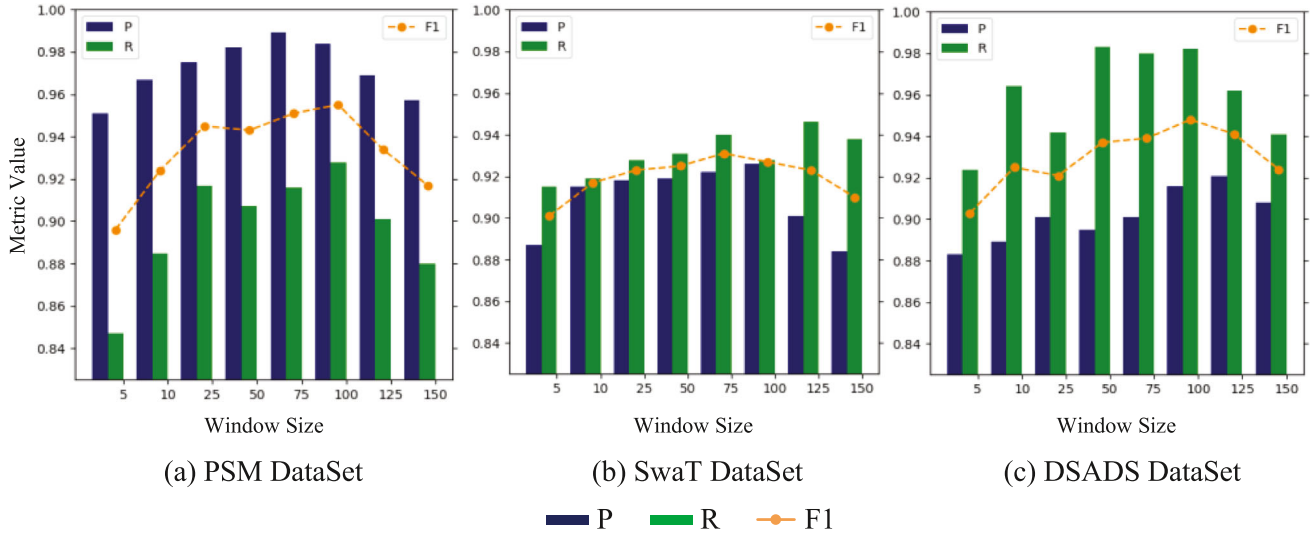


Fig. 7 The Impact of Different Window Sizes on the Anomaly Detection

Performance peaks at eight heads, after which additional heads lead to overfitting, redundancy, and higher computational overhead. Thus, eight attention heads were selected as the optimal configuration, as indicated by the grey dashed line.

Attention Embedding Dimension As shown in Fig. 8(e), the embedding dimension significantly influences the model's ability to represent features and compute attention scores, which are key to identifying dependencies within the input sequence. The attention mechanism, employed in both the encoder and decoder, benefits from larger embedding dimensions, as they allow for richer feature representations. Performance improves with increasing embedding dimensions, peaking at 1024, beyond which further increases result in diminishing returns and higher computational costs. Based on these findings, an embedding dimension of 1024 was selected as the optimal configuration.

Feature Compression Ratio The impact of the feature compression ratio on performance is shown in Fig. 8(f). A moderate compression ratio, such as 0.75, reduces redundancy in encoded features and enhances the efficiency of the decoding process. In contrast, lower compression ratios (e.g., 0.35 or 0.5) result in the loss of crucial information, significantly degrading performance. Higher compression ratios (e.g., 1.0) retain too much redundant information, adding unnecessary load to the decoder. Therefore, a compression ratio of 0.75 was chosen as the optimal setting to balance information preservation and processing efficiency.

5.9 Irregular sampling analysis

In some scenarios, time-series data is irregularly sampled, which is common in industrial production and sensor data. For example, equipment failures may result in irregular intervals between data points, or communication delays may lead to missing data. To assess the adaptability of the TADST under such conditions, we employed the following strategies, as summarized in Table 7:

Sampling Frequency Variation (SFV) This strategy adjusts the sampling frequency across time segments to simulate real-world variations. While the overall temporal structure is minimally affected, low-frequency intervals make it harder

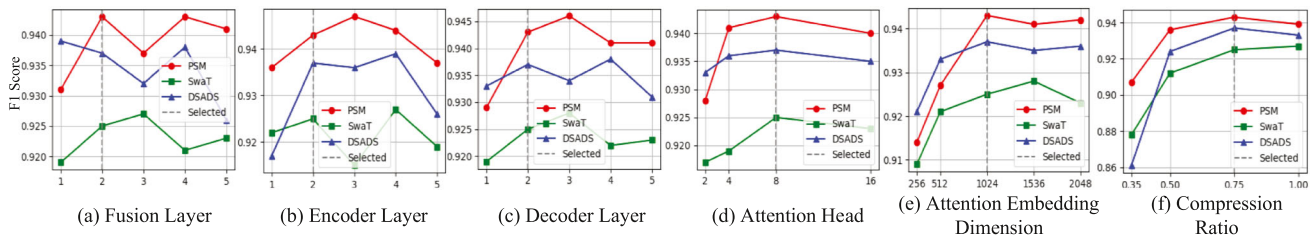


Fig. 8 TADST Parameter Sensitivity Analysis. The subfigures sequentially display the impact of spatio-temporal fusion layers, encoder layers, decoder layers, attention heads, attention embedding dimensions, and feature compression ratios on overall performance. Note that: the grey dashed lines indicate the optimal parameter configurations

Table 7 Performance of TADST under Different Sampling Strategies

Sampling Algorithm	Formula	PSM			SwaT		
		P	R	F1	P	R	F1
Original	$\{(t_i, x_i)\}_{i=1}^{L}$	0.982	0.907	0.943	0.919	0.931	0.925
SFV	$t_i = t_{k-1} + \frac{i}{fk}$	0.993	0.889	0.937	0.926	0.909	0.917
TIP	$t_i' = t_i + \delta t_i, \delta t_i \sim N(0, \sigma^2)$	0.976	0.891	0.932	0.939	0.923	0.931
NDS	$t_{i+1} = t_i + \Delta t_i, \Delta t_i \sim Exponential(\lambda)$	0.966	0.913	0.939	0.908	0.934	0.921
RMS	$\{(t_i, x_i)\|B_i = 1\}, B_i \sim Bernoulli(1-\rho)$	0.978	0.868	0.919	0.904	0.895	0.899

Note: "**Original**" indicates that no sampling algorithm is applied

for the model to detect high-frequency anomalies. Additionally, in sequences containing "persistent anomalies", the SFV leads to more anomaly points being missed. Recall rates on PSM and SWaT dropped by 1.984% and 2.363%, respectively, indicating robustness but with risks of missed detections in low-frequency intervals.

Time Interval Perturbation (TIP) Small-scale noise is added to timestamps to simulate irregular intervals. Since the perturbation is minor, data continuity is preserved. Model performance remains stable, demonstrating strong robustness to small temporal perturbations.

Non-uniform Distribution Sampling (NDS) Time intervals are generated using an exponential distribution to simulate variations in sampling density. This introduces significant variability, challenging the model's ability to capture temporal dependencies. On the PSM and SWaT datasets, F1 scores dropped by 0.424% and 0.432%, respectively. This demonstrates that the spatiotemporal feature fusion network allows the TADST to maintain overall performance.

Random Missing Sampling (RMS) Data points are randomly discarded using a Bernoulli distribution to simulate data loss. With a missing rate of $\rho = 0.3$, the loss of temporal information impacts anomaly detection in locally dependent regions. However, precision on PSM and SWaT dropped by only 0.407% and 1.632%, demonstrating stability under data loss.

5.10 Model interpretability

We analyze the model's attention focus on both normal and anomalous attributes and discuss the visualization results of global spatial dependencies across four datasets. The proposed model demonstrates strong interpretability by learning dependency structures through latent variables. The Graph Attention Influence Network (GAIN) represents multivariate time series as a graph structure, where the time series $\mathbf{x} \in \mathbb{R}^{L \times F}$ is transformed into a node set $V = \{v_1, \ldots, v_F\}$, with each node $v_i \in \mathbb{R}^L$, where F represents the total number of nodes and L denotes the feature dimension of each node.

When the sequence $\mathbf{x}$ contains no anomalous data, the GAIN effectively captures significant correlations between different attributes. These correlations are reflected in the attention heatmaps as darker blue regions, indicating higher attention weights among attributes when processing normal data, revealing a clear focus pattern. As shown in Fig. 9 (a-d), during the node feature update stage, each node selectively focuses on certain neighboring node features in addition to its own. This selective attention mechanism enhances the ability to aggregate information from neighboring nodes, improving its capacity to learn normal data patterns and its representation power for multivariate time series.

In contrast, when the sequence contains anomalous data, the correlations between attributes weaken significantly. This change is visualized in the attention heatmaps as lighter blue regions, indicating that the attention weights between anomalous attributes and other attributes decrease significantly, resulting in a more dispersed attention pattern. As shown in Fig. 9 (e-h), during the node feature update stage, the attention of each node to other features decreases noticeably, leading to a more scattered attention distribution. In the corresponding directed graph structures (Fig. 9 i-l), red nodes represent attributes containing anomalous sequences, while the color intensity of the directed edges reflects the connection strength between source and target nodes. From the subfigures, it can be observed that when red nodes act as source nodes, their influence on updating the features of other attributes is weaker. Similarly, when red nodes serve as target nodes, their neighboring nodes exert less influence on them.

This analysis reveals that the model effectively mitigates the negative impacts of anomalous data on normal sequences, ensuring that the feature update process for normal sequences remains largely unaffected by anomalous information. This capability validates the model's effectiveness in capturing anomalous patterns while maintaining robustness in processing normal data, providing strong support for its application in real-world anomaly detection tasks.

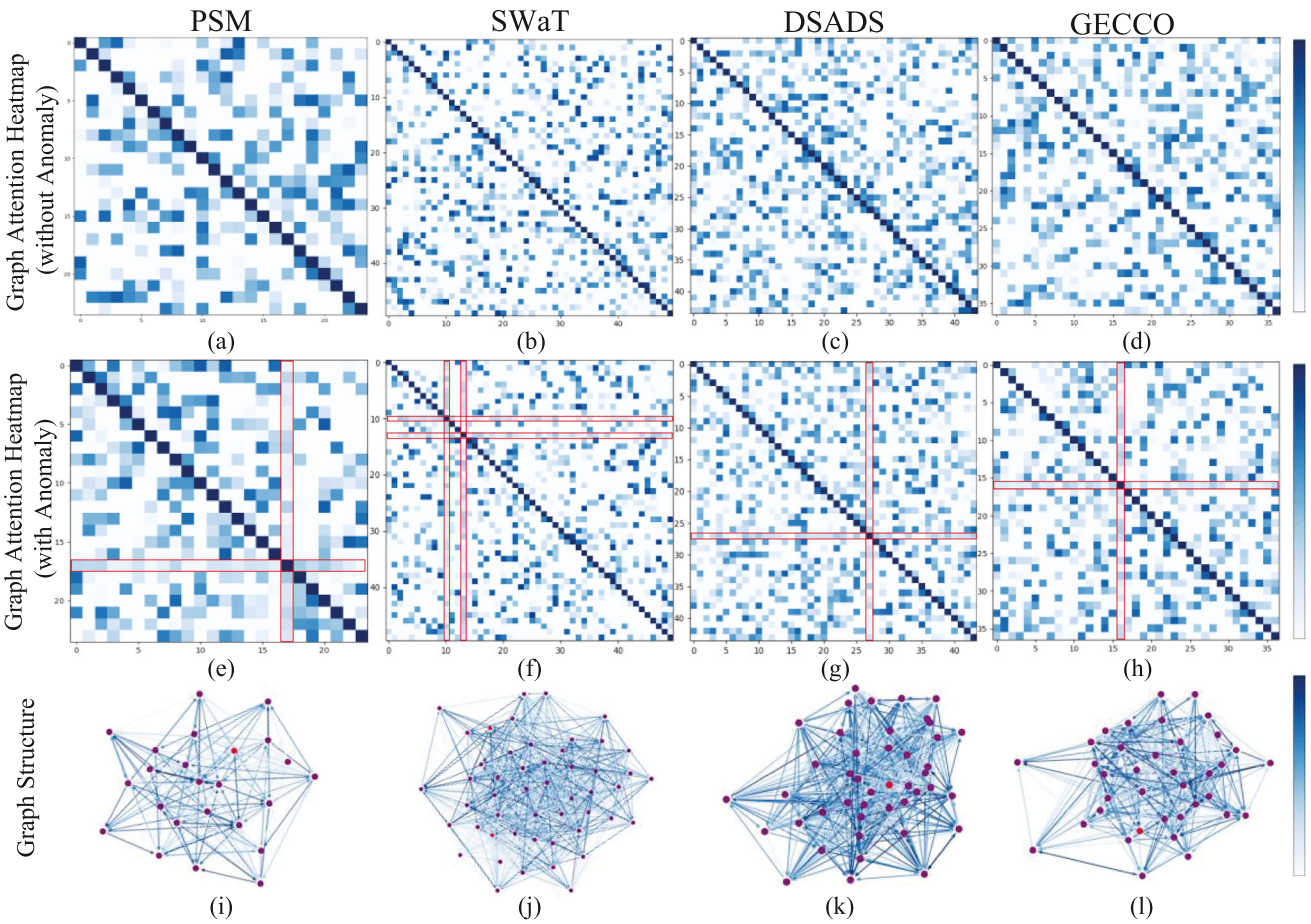


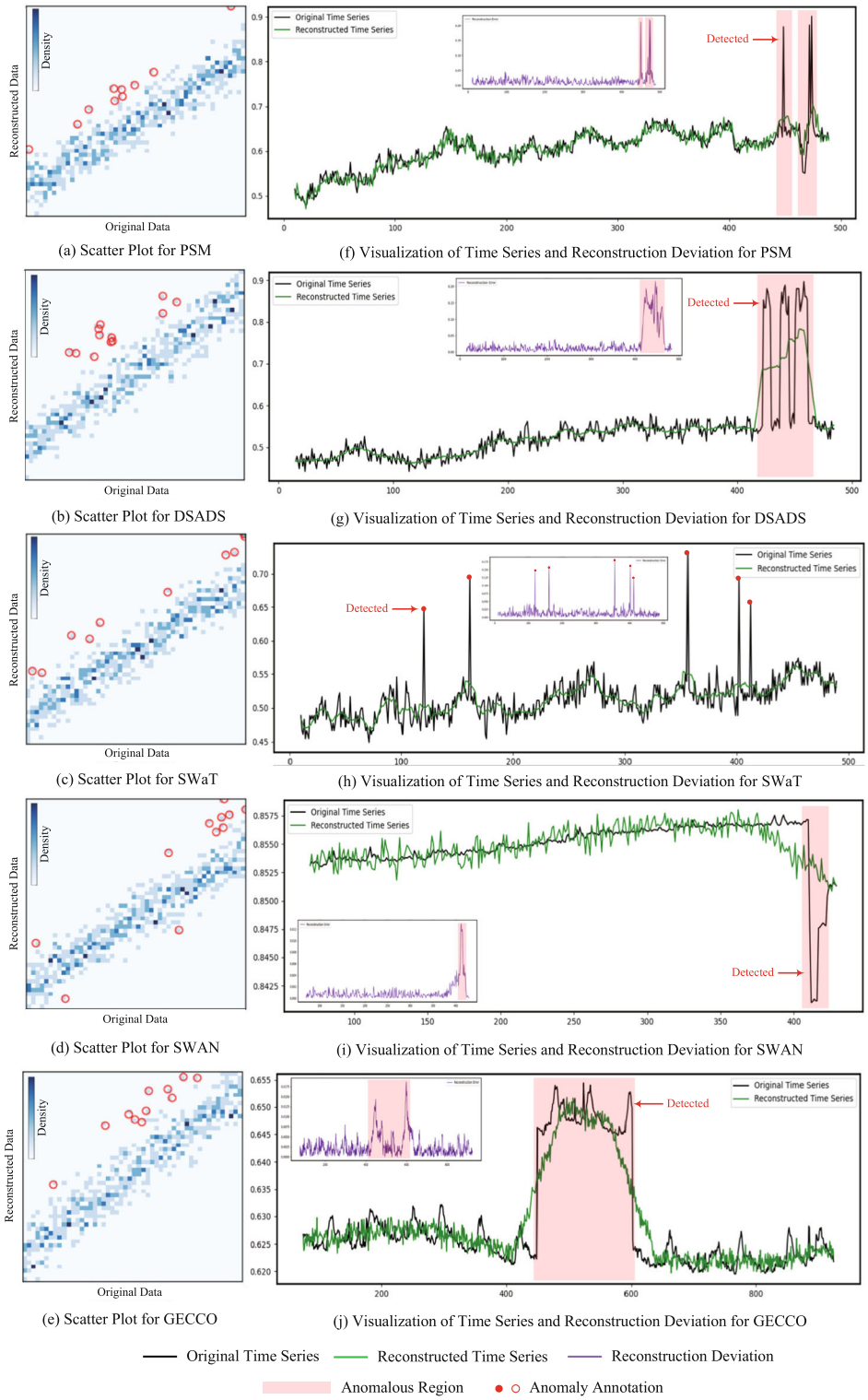
Fig. 9 Attention Heatmap and Directed Graph Structure. (a-d) Attention heatmap without abnormal attributes. (e-h) Attention heatmap with abnormal attributes. (i-l) Directed graph structure, with the red nodes indicating those containing anomalies

5.11 Visual analysis

Figure 10 illustrates the performance of TADST across different datasets: (1) The first column (a-e) represents the Anomaly Detection Mechanism (ADM) labeling potential anomaly points based on dynamic anomaly thresholds. (2) The second column (f-j) presents the reconstruction of the Time Series Reconstruction network (TSR) for the original time series, along with the reconstruction deviation between them. The detailed analysis is as follows:

Figure 10(a-e) demonstrate the reconstruction deviation and anomaly labeling results of TADST on different datasets. In these plots, the color intensity of the blue dots indicates the density of the data points, the x-axis represents observed values (including anomalies), and the y-axis represents reconstructed values (i.e., predicted values). The closer the data points are to the main diagonal, the smaller the reconstruction deviation, indicating that the original data conforms more closely to the normal distribution learned by TADST. The results show that when data points deviate from normal patterns, leading to reconstruction deviations exceeding the corresponding thresholds, the ADM effectively identifies these potential anomalies (marked in red) without generating false positives. Furthermore, these subplots reveal that potential anomalies are generally located in regions of lower data density.

Figure 10(f-j) illustrate the original time series (black), the reconstructed time series (green), and the embedded plots of reconstruction deviations (purple). In these subplots, anomalies primarily include "isolated point anomalies", "local anomalies", and "persistent anomalies": (***i***) In the PSM, DSADS, and SWAN samples (Fig. 10 f, g, i), small-scale local anomalies are observed, causing abrupt changes in specific regions that quickly return to normal. The embedded plots reveal that the reconstruction deviations in non-anomalous regions remain low, while in local anomaly regions, TSR is minimally affected by potential anomalies. This is due to the network's spatiotemporal feature fusion method, which captures long-range dependencies and global correlations among different attributes. Consequently, even when specific attributes exhibit anomalies in certain time periods, the network effectively leverages spatiotemporal



(a) Scatter Plot for PSM
(f) Visualization of Time Series and Reconstruction Deviation for PSM
(b) Scatter Plot for DSADS
(g) Visualization of Time Series and Reconstruction Deviation for DSADS
(c) Scatter Plot for SWaT
(h) Visualization of Time Series and Reconstruction Deviation for SWaT
(d) Scatter Plot for SWAN
(i) Visualization of Time Series and Reconstruction Deviation for SWAN
(e) Scatter Plot for GECCO
(j) Visualization of Time Series and Reconstruction Deviation for GECCO

Fig. 10 Case Studies on the PSM, DSADS, SWaT, SWAN, and GECCO Datasets. (a-e) illustrate the anomalies detected by ADM on the five datasets based on dynamic anomaly thresholds. (f-j) present the visualizations of time series reconstruction by TSR for the samples from the five datasets, with each subfigure consisting of the Original Time Series, Reconstructed Time Series, and Reconstruction Deviation

information to mitigate the interference caused by anomalies. (***ii***) In the SWaT sample (Fig. 10 h), a few isolated point anomalies are detected. Compared to local anomalies, TADST is less affected by potential point anomalies. Given the lower prediction difficulty of point anomalies, the reconstructed values can be utilized as corrected values for these anomalies, enhancing TADST's versatility. (***iii***) In the GECCO sample (Fig. 10 j), a sequence of length 1000 is reconstructed. Persistent anomalies spanning long durations significantly increase prediction difficulty. Although the sparse attention mechanism and temporal convolutional network effectively capture global sequence information, the model demonstrates slightly reduced performance in handling "persistent anomalies" compared to "isolated point anomalies" and "local anomalies". However, the embedded plot shows that despite decreased prediction performance in the middle of the anomalous range (e.g., timestamps 500-550), TADST still amplifies reconstruction deviations in the leading (e.g., timestamps 450-500) and trailing (e.g., timestamps 550-600) segments of the anomalous region by learning the normal data distribution.

6 Conclusion

This paper introduces a novel anomaly detection model (TADST), which achieves state-of-the-art performance across five benchmark datasets. By dynamically decomposing time series, the model effectively reduces false positives caused by non-anomalous transient signals. The Feature Fusion Network captures temporal dependencies and global attribute correlations, resulting in a more comprehensive feature representation. Furthermore, the Feature Compression block eliminates redundancies in the encoded features, enabling the model to learn the normal distribution and behavioral patterns of the data, while accurately predicting normal behavior for anomalous instances. Lastly, the Anomaly Detection Mechanism dynamically adjusts thresholds, further reducing false alarms.

In conclusion, TADST not only addresses robustness challenges related to redundant information and threshold configuration but also enhances interpretability in anomaly localization. Future work will explore reinforcement learning-based adaptive optimization of anomaly detection strategies.

Author Contributions Bin Yang contributed to the preliminary research, model design, and manuscript writing. Tinghuai Ma contributed to the conceptualization and methodology. Huan Rong, Xuejian Huang, Yubo Wang, Bowen Zhao, and Chaoming Wang contributed to the ablation experiments and comparative experiments.

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Declarations

Competing Interests The authors have no conflict of interest.

Ethical and informed consent Not applicable

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